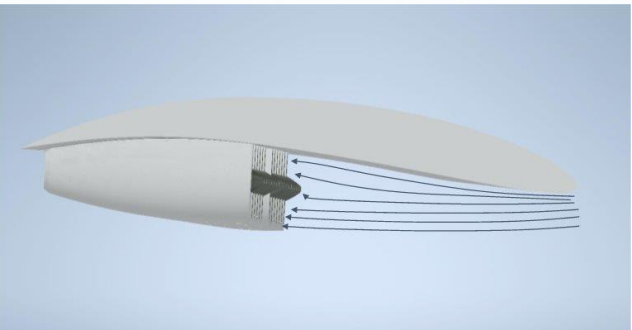


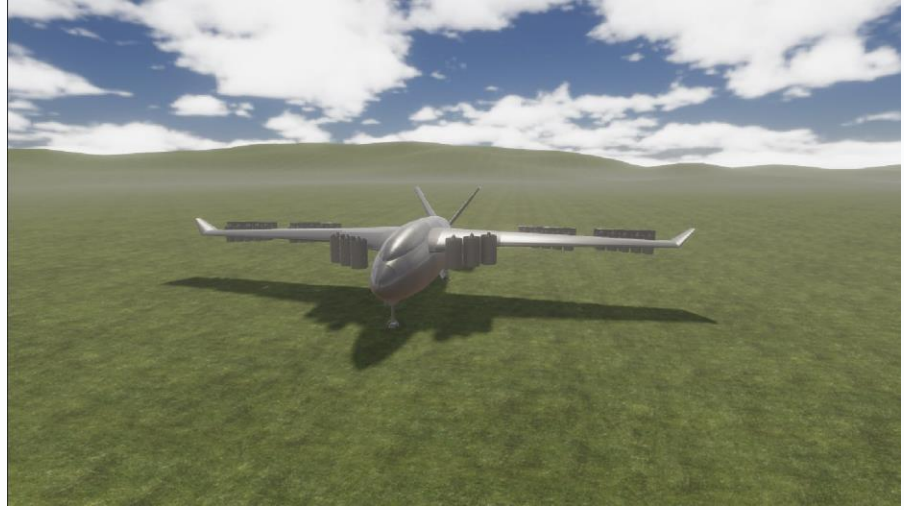


1. Introduction

UAM/eVTOL aircraft often use distributed EDF propulsion for compact integration and potential noise benefits. But EDF placement near an airfoil changes local flow, drag, and wake behavior. This study examines those aerodynamic trade-offs to support better eVTOL design.



Maldonado-Hicks Airfoil

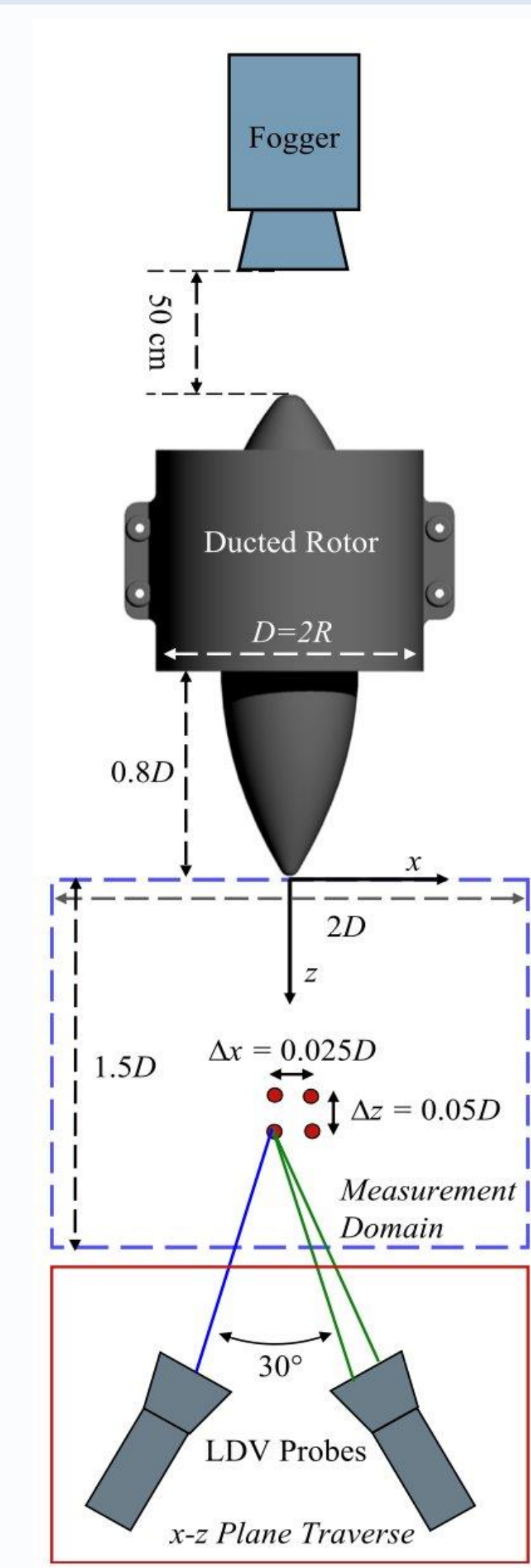


2. Objective / Research Question

Research Question:
How does EDF placement near the aft section of an airfoil affect aerodynamic drag, thrust-related performance, and wake behavior?

- Objectives:**
- Study EDF–airfoil aerodynamic interaction
 - Vary EDF position relative to the airfoil
 - Vary inlet spacing / trailing-edge spacing
 - Evaluate changes in drag and wake structure
 - Compare performance trends across configurations
 - Provide guidance for future eVTOL/UAM airframe-propulsion integration

3. Methods / Approach

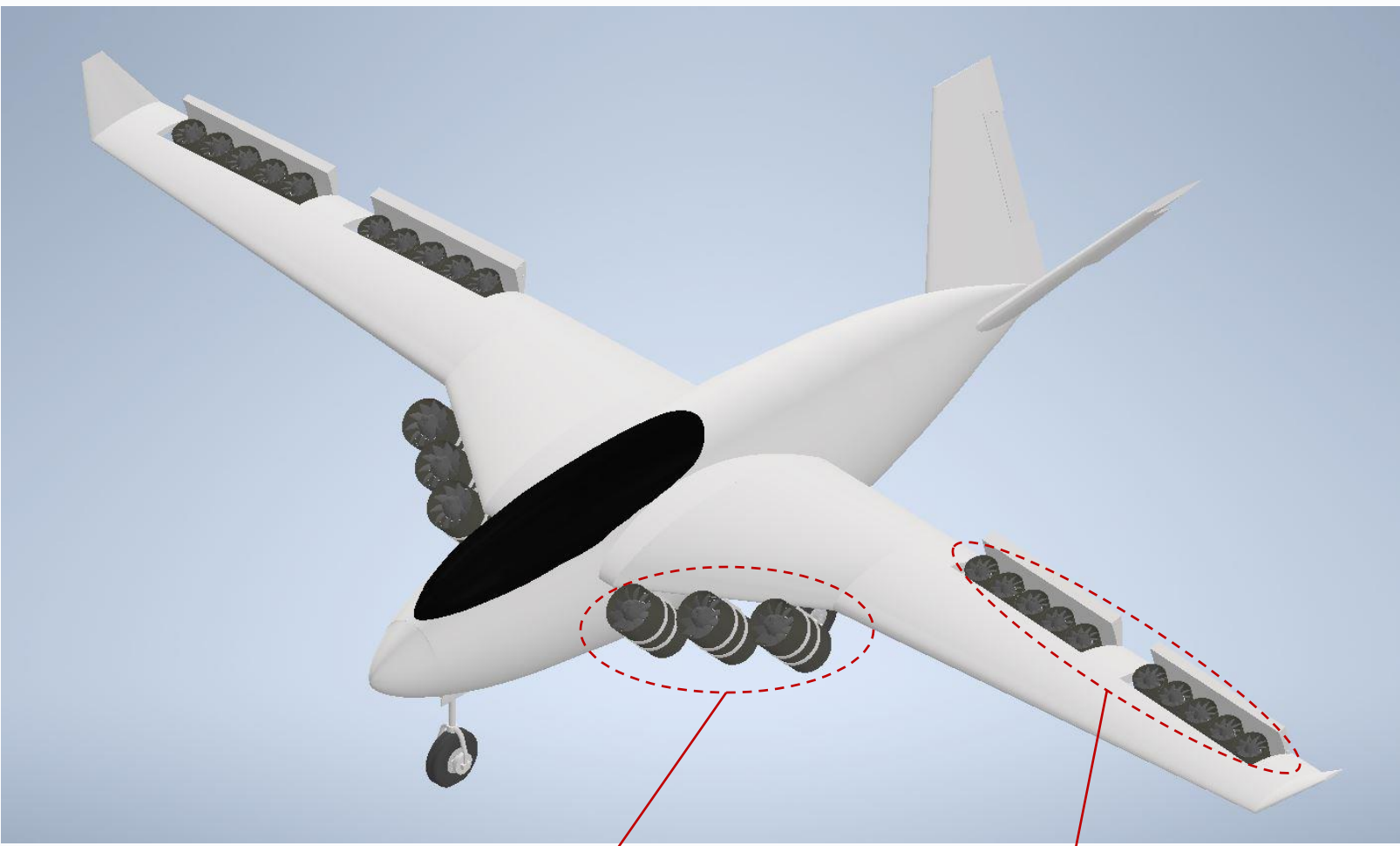


- CFD Approach:**
- Modeled airflow around an airfoil with an integrated horizontal EDF
 - Examined multiple EDF placement locations near the aft airfoil region
 - Varied inlet spacing / trailing-edge-to-inlet distance
 - Evaluated aerodynamic quantities including drag, flow acceleration, and wake structure
 - Compared how fan placement modified local pressure and downstream flow behavior

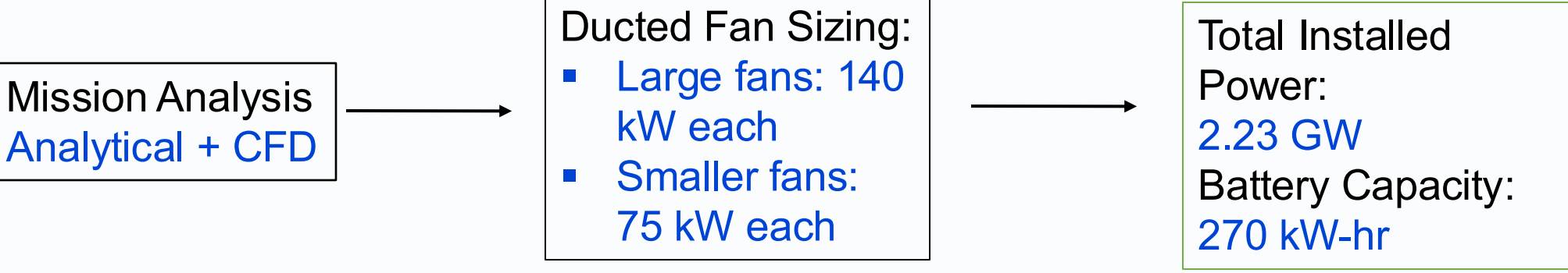
- Experimental / Validation Approach:**
- Used wind tunnel testing to examine key aerodynamic trends
 - Compared wake features and performance tendencies with CFD predictions
 - Identified consistent system-level effects of EDF–airfoil interaction

4. Key Figure / System Overview

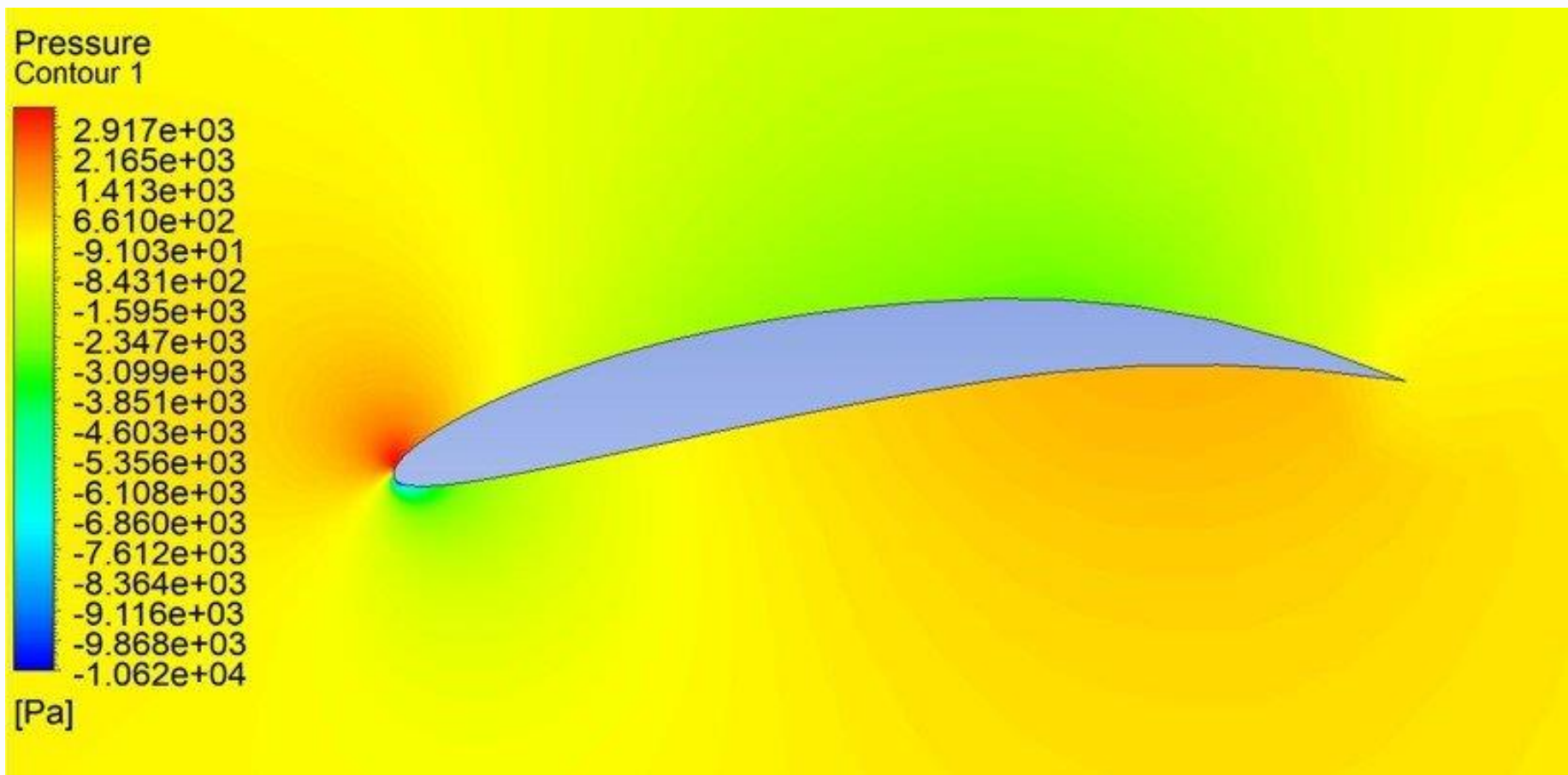
Ducted Fan Studies: eVTOL context



6 large inboard fans 20 smaller flap fans



Maldonado-Hicks Airfoil

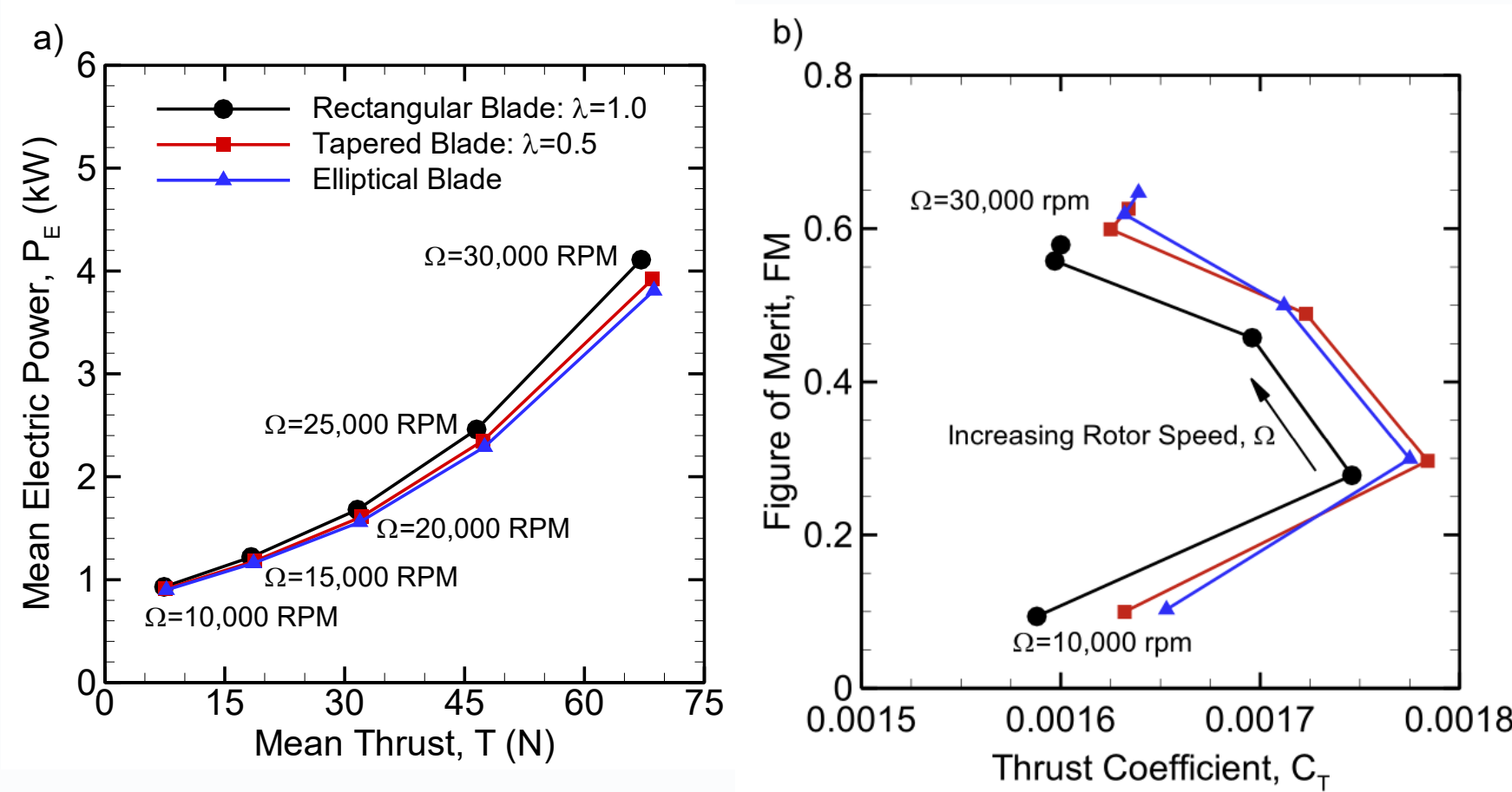


Maldonado-Hicks Airfoil Design The Maldonado-Hicks airfoil was developed to blend the ducted fan with the undersurface of the airfoil in order to improve aerodynamic performance. The airfoil geometry was defined using 5th-order polynomial functions with boundary conditions that control the shape near the trailing edge and fan inlet. This blended configuration reduces interference and form drag, improves flow guidance into the fan, and helps achieve more optimal EDF-airfoil integration for future eVTOL applications.

5. Results / Data

Results: Impact of Blade Shape
Ducted Fan Performance

- Ducted fans are power hungry!
- Elliptical fan blades improve the FM and C_T performance by up to ~12% and ~2.3%



Performance Parameters

Figure of merit:

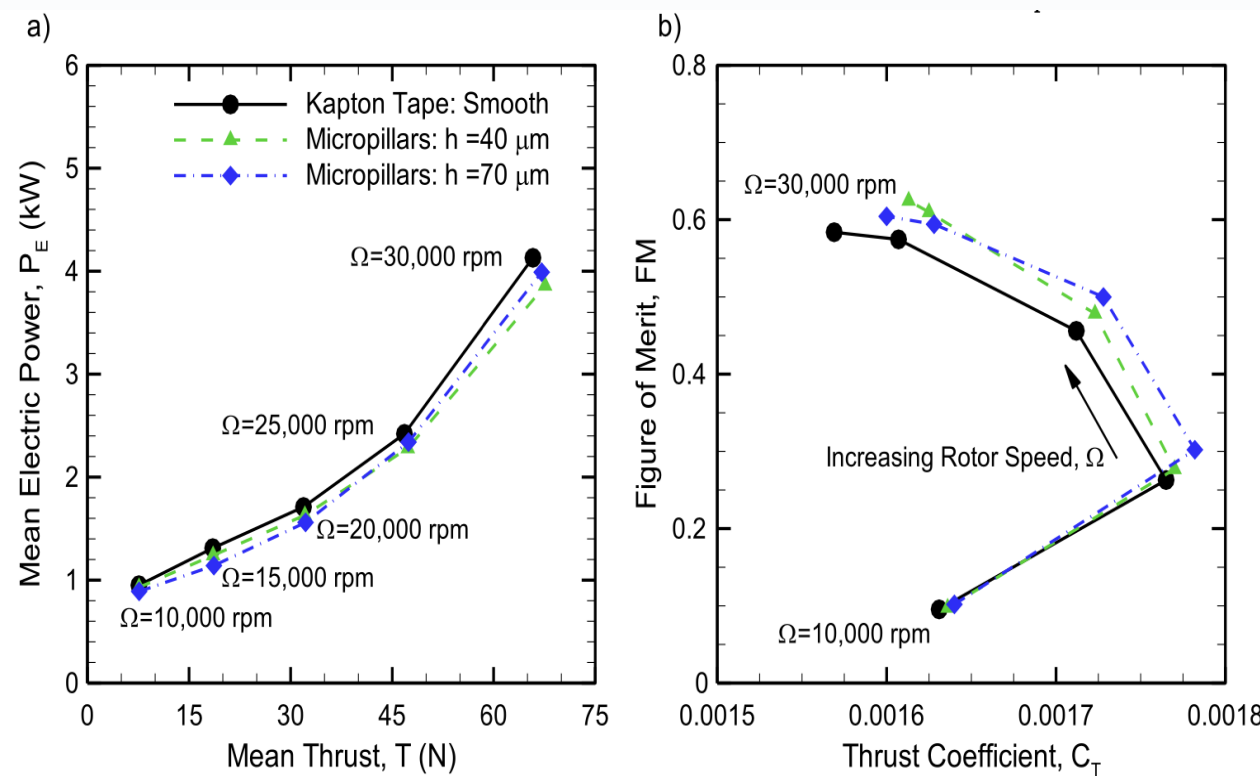
$$FM = \frac{P_{ideal}}{P_{actual}} = \frac{1}{2} \sqrt{\frac{T^3}{\rho A \sigma_d}}$$

Thrust coefficient:

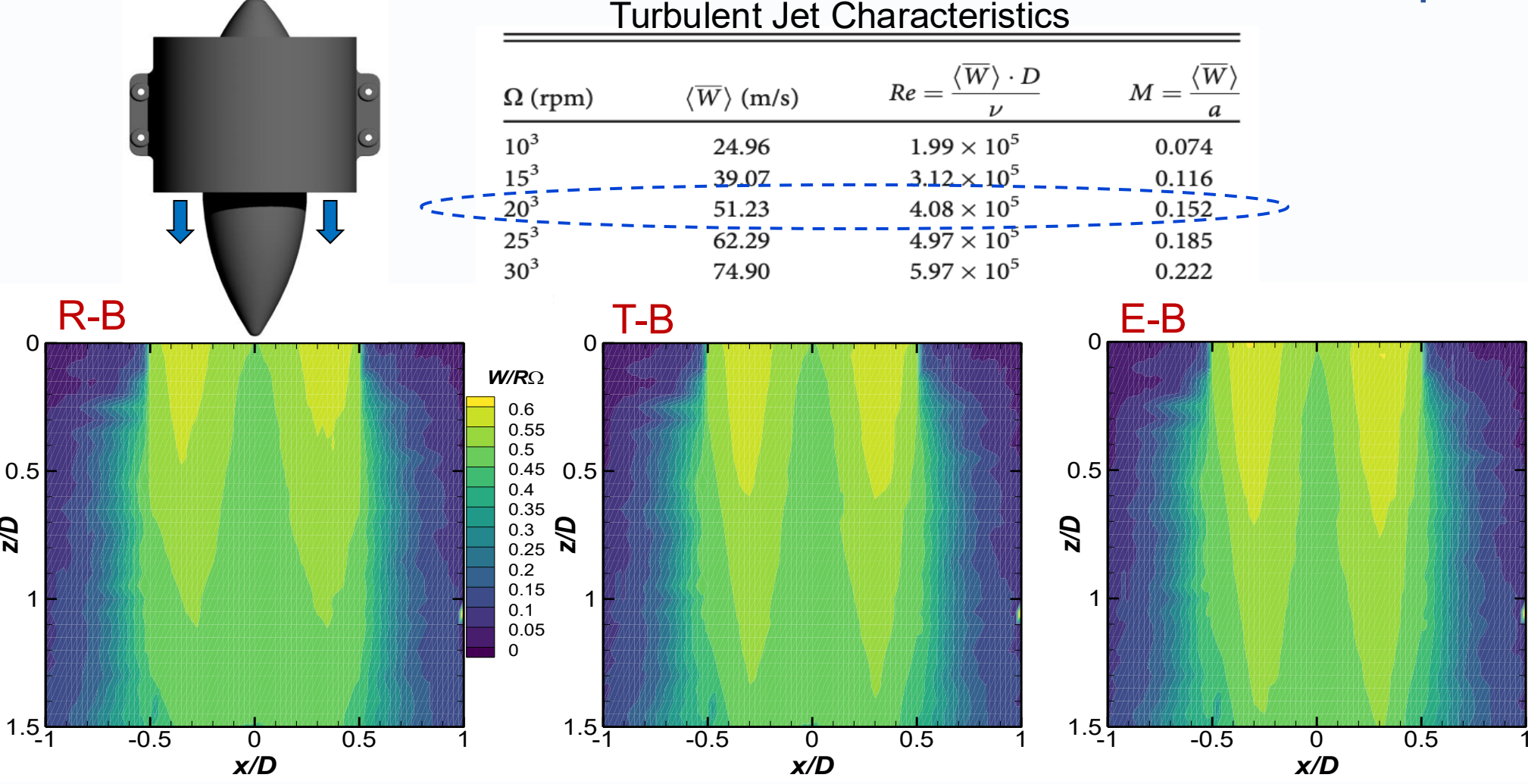
$$C_T = \frac{T}{\rho A \Omega^2 R^2}$$

Results: Impact of 'Sharkskin' Surface
Ducted Fan Performance

- Micropillars with $h = 40 \mu m$ improve the FM and C_T performance by up to ~7% and ~2.7%
- Performance depends on fan speed and Re , which changes the variable δ/h



Results: Impact of Blade Shape
Time-Averaged Streamwise Flow, $W/R\Omega$ at $\Omega = 20,000$ rpm



Result highlights:

- EDF presence significantly altered local airflow around the airfoil
- Drag was sensitive to fan position and inlet spacing
- Some configurations increased drag due to stronger aerodynamic interference
- Other configurations improved local flow guidance into the fan
- Wake structure changed with placement, affecting down stream flow quality
- Wind tunnel observations supported the main CFD trends

6. Discussion / Interpretation

The results indicate that **EDF-airfoil integration strongly affects aerodynamic behavior**. Placement changes the local pressure field, modifies flow entering the fan, and alters the downstream wake. This means propulsion and airframe cannot be designed independently in future eVTOL concepts. Instead, **fan location, inlet spacing, and surrounding airfoil geometry must be optimized together** to reduce aerodynamic penalties and improve overall System efficiency.

7. Conclusion

EDF placement near an airfoil has a significant effect on aerodynamic performance. Both **fan position** and **inlet spacing** influence drag, local flow behavior, and wake development. The study shows that careful propulsion-airframe integration can reduce penalties and improve system performance. These findings provide useful design guidance for future **safer, quieter, and more efficient UAM/eVTOL vehicles**.

8. Future Direction

- Test additional EDF placement configurations
- Study different airfoil shapes and thicknesses
- Include acoustic / noise effects
- Expand from 2D or section-level analysis to full wing models
- Extend to full-vehicle eVTOL conceptual design
- Investigate control, stability, and mission-level trade-offs

Maldonado & Fernandes (2024), *Physics of Fluids* — blade shape effects in ducted rotors.
Maldonado, Fernandes & Mallory (2025), *Aerospace Science and Technology* — sharkskin-inspired surface effects on ducted fan jets.
Hicks et al. (2023), AIAA Aviation Forum — underwing FanFoil eVTOL concept.

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